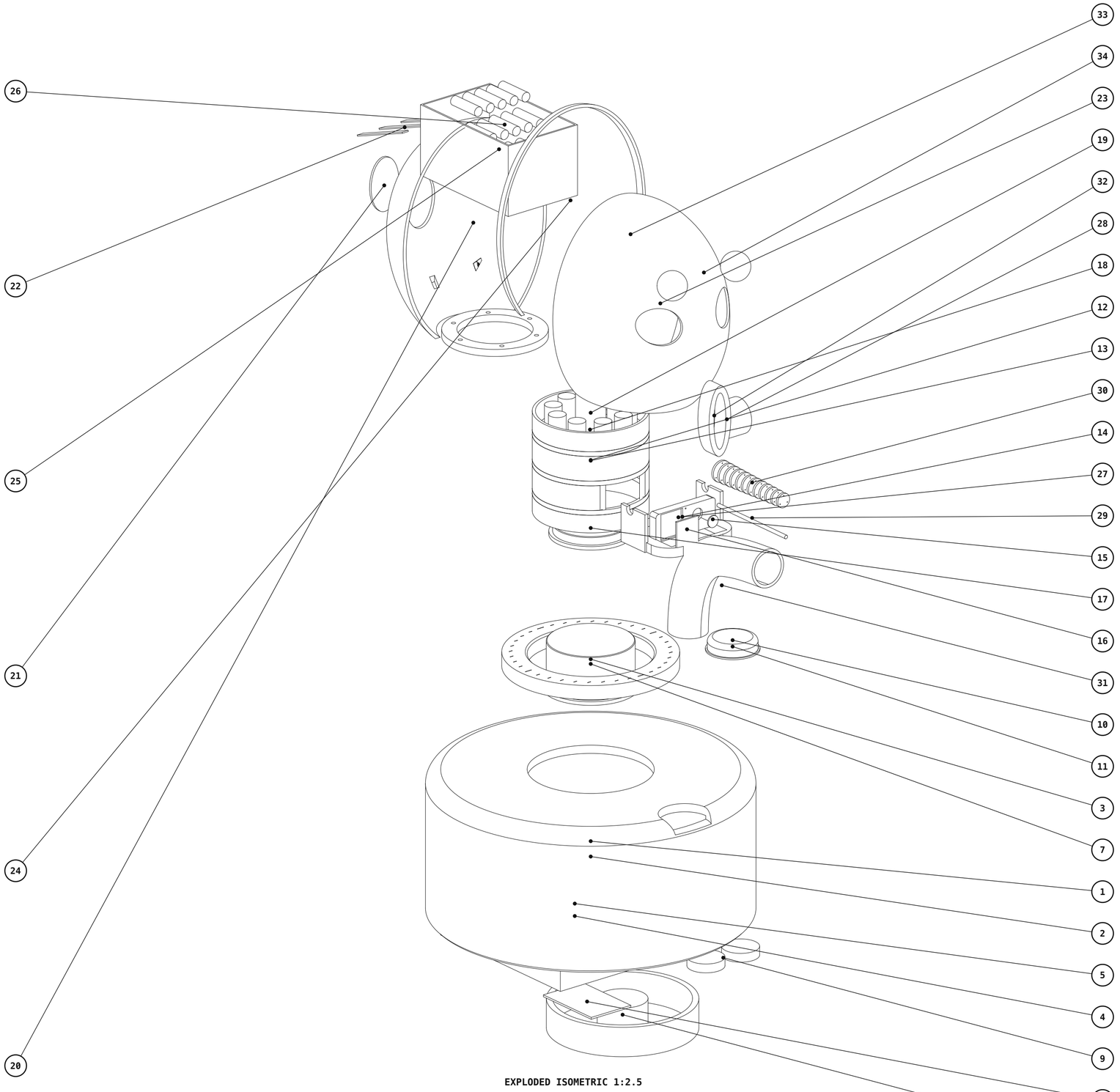
WIP - ENGINEERING REVIEW ONLY - NOT RELEASED FOR FABRICATION



EXPLODED ISOMETRIC 1:2.5

ITEM	PART NO.	DESCRIPTION	QTY	MATERIAL	KIND	
1	SBL-MU-019	Cradle shell	1	Aluminium 6061-T6	custom	
2	SBL-MU-020	Ballast ring	1	Stainless 304L, ASTM A276	custom	
3	SBL-MU-021	Cradle top ring	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
4	SBL-MU-022	Heat sink	1	Copper vapour chamber + Al 6063 fins	custom	
5	SBL-MU-B01	Compute module (Jetson Thor class)	1	FR-4 PCB, 1.6 mm, black soldermask	bought	
6	SBL-MU-B02	Fan 120 mm low-rpm	1	Bought component (datasheet envelope	bought	
7	SBL-MU-B03	Yaw strain-wave joint	1	Bought component (datasheet envelope	bought	
8	SBL-MU-B04	Main safety MCU board	1	FR-4 PCB, 1.6 mm, black soldermask	bought	
9	SBL-MU-B05	Disc pump ×2 (suction, lips)	2	Bought component (datasheet envelope	bought	
10	SBL-MU-023	STOP mushroom	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
11	SBL-MU-024	STOP light ring	1	Optical silicone light guide	custom	
12	SBL-MU-016	Neck column	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
13	SBL-MU-017	Neck rings ×3	3	3Y-TZP zirconia, black	custom	
14	SBL-MU-018	Sensor collar	1	Ti-6Al-4V ELI, ASTM F136	custom	
15	SBL-MU-018a	Thermal window Ø10	1	Germanium, AR-coated 8–14 µm	custom	
16	SBL-MU-018b	Radar radome 22×22	1	PEEK (Victrex 450G, black)	custom	
17	SBL-MU-B06	Neck pitch/roll joint	1	Bought component (datasheet envelope	bought	
18	SBL-MU-B07	Tongue tendon drive pack (9 spools + 1 axial)	10	Bought component (datasheet envelope	bought	
19	SBL-MU-B08	Throat channel tube	1	Medical LSR, Shore A20–A40	moulded	
20	SBL-MU-003	Occipital chassis	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
21	SBL-MU-005	Occipital window	1	C-plane sapphire, AR both sides	custom	
22	SBL-MU-003a	Cooling inlays ×3	3	3Y-TZP zirconia, black	custom	
23	SBL-MU-002	Front skull shell	1	PEEK 30% CF (Victrex 450CA30)	custom	
24	SBL-MU-004	Skin clamp band	1	Ti-6Al-4V ELI, ASTM F136	custom	
25	SBL-MU-006	Actuator bank frame	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
26	SBL-MU-B09	Micro linear actuator ×15	15	Bought component (datasheet envelope	bought	
27	SBL-MU-007	Mandible frame	1	Ti-6Al-4V ELI, ASTM F136	custom	
28	SBL-MU-008	Jaw drive housing	1	Ti-6Al-4V, ASTM B348 Gr 5	custom	
29	SBL-MU-009	Tongue backbone	1	Nitinol superelastic tube	custom	
30	SBL-MU-010	Tongue discs ×12	12	PEEK (Victrex 450G, black)	custom	
31	SBL-MU-012	Oral sleeve (removable)	1	Medical LSR, Shore A20–A40	moulded	
32	SBL-MU-013	Lip core bladders	1	Medical LSR, Shore A20–A40	moulded	
33	SBL-MU-025	Palate speaker chamber	1	PEEK (Victrex 450G, black)	custom	
34	SBL-MU-014	Eye ×2	2	PMMA cornea over glass core	bought	

1. Explode offsets are illustrative; items balloon to the BOM.
2. Bought components are datasheet envelopes (kind = bought).

SABLE ROBOTICS — **SBL-MU**

EXPLODED ASSEMBLY DRAWING · Sable Muse — **REV A**

MATERIAL	FINISH	SCALE
see BOM	see BOM	
GENERAL TOLERANCE	SURFACE	MASS
ISO 2768-mK · UNITS mm · ANGLES °	see Ra callouts; unmarked edges 0.6 × 45°	—
THIRD ANGLE PROJECTION	SHEET 2 / 6	A2 · 2026-11-08 · parts/cad/src